Math 54 — Week 6

Linearly independent sets and bases; dimension and rank (Lay 4.3, 4.5)

§1 Facts used from Weeks 1–5

(W1) In a dependent set $\{\mathbf{v}_1, \dots, \mathbf{v}_p\}$ with $\mathbf{v}_1 \neq \mathbf{0}$, some $\mathbf{v}_j$ is a combination of $\mathbf{v}_1, \dots, \mathbf{v}_{j-1}$. More vectors than entries in $\mathbb{R}^n$ are dependent.

(W2) Columns of A independent $\iff$ pivot in every column. $A\mathbf{x} = \mathbf{0}$ has a nontrivial solution $\iff$ a free variable. A homogeneous system with more unknowns than equations has a nontrivial solution.

(W3) Row operations do not change the solution set of $A\mathbf{x} = \mathbf{0}$.

(W4) Subspaces; Span is a subspace; $\operatorname{Nul} A$, $\operatorname{Col} A$, $\operatorname{Row} A$. The parametric vector form of $A\mathbf{x} = \mathbf{0}$ gives a spanning set for $\operatorname{Nul} A$, one vector per free variable, and that set is independent.

§2 This week

§2.1 Linear independence in a vector space; bases (Lay 4.3)

- The definitions carry over unchanged: $\{\mathbf{v}_1, \dots, \mathbf{v}_p\}$ in V is **linearly independent** if $c_1\mathbf{v}_1 + \cdots + c_p\mathbf{v}_p = \mathbf{0}$ only for $c_1 = \cdots = c_p = 0$. The theorem (W1) about some $\mathbf{v}_j$ being a combination of the earlier vectors holds in any vector space, with the same proof.
- In $\mathbb{P}_2$, $\{1,\ t,\ t^2\}$ is independent: $c_0 + c_1 t + c_2 t^2$ is the zero polynomial only if all coefficients are 0. $\{\sin t \cos t,\ \sin 2t\}$ is dependent in the space of functions, since $\sin 2t = 2 \sin t \cos t$.

The question. A spanning set may contain redundant vectors. What is the smallest kind of set that still spans, and how do we find one for $\operatorname{Nul} A$ and $\operatorname{Col} A$?

Definition. A **basis** for a subspace H of V is a set $\mathcal{B} = \{\mathbf{b}_1, \dots, \mathbf{b}_p\}$ in H that is linearly independent and spans H.

- The columns $\mathbf{e}_1, \dots, \mathbf{e}_n$ of I_n form the **standard basis** of $\mathbb{R}^n$. $\{1, t, \dots, t^n\}$ is the standard basis of $\mathbb{P}_n$.
- The columns of an invertible $n \times n$ matrix form a basis of $\mathbb{R}^n$: by the invertible matrix theorem they are independent and span $\mathbb{R}^n$.

Theorem (Spanning set theorem)
Let $S = \{\mathbf{v}_1, \dots, \mathbf{v}_p\}$ and $H = \operatorname{Span} S$. (a) If some $\mathbf{v}_k$ is a combination of the others, removing it from S does not change the span. (b) If $H \neq \{\mathbf{0}\}$, some subset of S is a basis for H.

Proof. 1. (a) Write $\mathbf{v}_k$ as a combination of the other vectors: $\mathbf{v}_k = \sum_{i \neq k} a_i \mathbf{v}_i$. In any combination $\sum c_i \mathbf{v}_i$, replace $\mathbf{v}_k$ by this expression; the result uses only the other vectors.

2. (b) If S is independent it is a basis. Otherwise some vector is a combination of the others (W1); remove it by (a). Repeat. The process stops, at the latest when one nonzero vector remains, which is independent.

□

A basis is a spanning set with nothing removable, and an independent set to which nothing in H can be added without losing independence.

A basis for $\operatorname{Nul} A$: the vectors from the parametric vector form of $A\mathbf{x} = \mathbf{0}$ (W4).

Theorem (A basis for $\operatorname{Col} A$)
The pivot columns of A form a basis for $\operatorname{Col} A$.

The idea

Row reduction changes the columns, so $\operatorname{Col} A$ and $\operatorname{Col} U$ are usually different spaces. But it does not change *which combinations of the columns are zero*: a relation $c_1\mathbf{a}_1 + \cdots + c_n\mathbf{a}_n = \mathbf{0}$ is a solution of $A\mathbf{x} = \mathbf{0}$, and A and its echelon form U have the same solutions (W3). In U it is visible that the pivot columns are independent and every other column is a combination of them. The same relations therefore hold among the columns of A.

Proof. 1. Let B be the reduced echelon form of A. $A\mathbf{x} = \mathbf{0}$ and $B\mathbf{x} = \mathbf{0}$ have the same solutions, so a list of weights gives $\mathbf{0}$ on the columns of A exactly when it gives $\mathbf{0}$ on the same columns of B.

2. The pivot columns of B are distinct standard vectors $\mathbf{e}_i$, so they are independent; each non-pivot column of B is a combination of the pivot columns before it.

3. By step 1, the pivot columns of A are independent, and each non-pivot column of A is the same combination of pivot columns of A. By the spanning set theorem the non-pivot columns can be removed without changing $\operatorname{Col} A$.

□

Worked example.

$$A = \begin{bmatrix} 1 & 4 & 0 & 2 & -1 \\ 3 & 12 & 1 & 5 & 5 \\ 2 & 8 & 1 & 3 & 2 \\ 5 & 20 & 2 & 8 & 8 \end{bmatrix} \longrightarrow \text{RREF} \begin{bmatrix} 1 & 4 & 0 & 2 & 0 \\ 0 & 0 & 1 & -1 & 0 \\ 0 & 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 & 0 \end{bmatrix}.$$

1. Pivot columns 1, 3, 5. Basis for $\operatorname{Col} A$: $\mathbf{a}_1 = (1, 3, 2, 5)$, $\mathbf{a}_3 = (0, 1, 1, 2)$, $\mathbf{a}_5 = (-1, 5, 2, 8)$ (columns of A, not of the echelon form).
2. x_2, x_4 free; $x_1 = -4x_2 - 2x_4$, $x_3 = x_4$, $x_5 = 0$. Basis for $\operatorname{Nul} A$: $(-4, 1, 0, 0, 0)$, $(-2, 0, 1, 1, 0)$.
3. Basis for $\operatorname{Row} A$: the nonzero rows of an echelon form, here $(1, 4, 0, 2, 0)$, $(0, 0, 1, -1, 0)$, $(0, 0, 0, 0, 1)$. (Row operations replace rows by combinations of rows and can be undone, so they do not change the row space; the nonzero rows of an echelon form are independent, since each has a leading entry in a column where the rows below have zeros.)

§2.2 Dimension (Lay 4.5)

The question. A subspace has many bases. Do they all have the same number of vectors? If so, that number measures the size of the subspace.

Theorem (More vectors than a basis are dependent)
If V has a basis $\{\mathbf{b}_1, \ldots, \mathbf{b}_n\}$, then any set of more than n vectors in V is linearly dependent.

The idea

Write each of the vectors $\mathbf{u}_1, \ldots, \mathbf{u}_p$ $(p > n)$ in terms of the basis. Finding a dependence relation among the $\mathbf{u}$'s then amounts to solving a homogeneous linear system with n equations (one per basis vector) and p unknowns (one weight per $\mathbf{u}_j$). More unknowns than equations always gives a nontrivial solution (W2).

Proof. 1. Write $\mathbf{u}_j = a_{1j}\mathbf{b}_1 + \cdots + a_{nj}\mathbf{b}_n$ for $j = 1, \ldots, p$, and let $M = (a_{ij})$, an $n \times p$ matrix.

2. Since $p > n$, $M\mathbf{c} = \mathbf{0}$ has a nontrivial solution $\mathbf{c} = (c_1, \ldots, c_p)$.
3. Then
$$\sum_j c_j \mathbf{u}_j = \sum_j c_j \sum_i a_{ij}\mathbf{b}_i = \sum_i (\sum_j a_{ij}c_j)\mathbf{b}_i = \sum_i (M\mathbf{c})_i \mathbf{b}_i = \mathbf{0},$$
with the c_j not all 0.

□

Theorem (All bases have the same size)
If V has a basis of n vectors, every basis of V has exactly n vectors.

Proof. If $\mathcal{C}$ is another basis, it is independent, so it has at most n vectors by the previous theorem. Exchanging the roles of the two bases, n is at most the size of $\mathcal{C}$. □

Definition. If V is spanned by a finite set, V is **finite-dimensional**, and $\dim V$ is the number of vectors in any basis. $\dim\{\mathbf{0}\} = 0$. Otherwise V is **infinite-dimensional**.

- $\dim \mathbb{R}^n = n$; $\dim \mathbb{P}_n = n + 1$; the space of all polynomials is infinite-dimensional.
- Subspaces of $\mathbb{R}^3$ by dimension: 0: $\{\mathbf{0}\}$; 1: lines through $\mathbf{0}$; 2: planes through $\mathbf{0}$; 3: $\mathbb{R}^3$ itself.

Theorem (Subspaces; extending to a basis)
If H is a subspace of a finite-dimensional V, any independent set in H can be extended to a basis of H, and $\dim H \leq \dim V$.

Proof. If an independent set S in H does not span H, pick $\mathbf{u} \in H$ outside $\operatorname{Span} S$; then $S \cup \{\mathbf{u}\}$ is still independent (in a relation, the weight on $\mathbf{u}$ must be 0, else $\mathbf{u} \in \operatorname{Span} S$; then all weights are 0). Repeat. By the theorem above the size never exceeds $\dim V$, so the process stops, with a basis of H. □

Theorem (Basis theorem)
Let V have dimension $p \geq 1$. Any p linearly independent vectors in V form a basis. Any p vectors that span V form a basis.

Proof. An independent set of p vectors extends to a basis, which has exactly p vectors, so nothing was added. A spanning set of p vectors contains a basis (spanning set theorem), which has exactly p vectors, so nothing was removed. □

So when you already know the dimension, you only need to check *one* of the two conditions.

§2.3 Rank (Lay 4.5)

- $\dim \operatorname{Nul} A$ = number of free variables. $\dim \operatorname{Col} A$ = number of pivot columns. $\dim \operatorname{Row} A$ = number of pivots (the nonzero rows of an echelon form).
- The **rank** of A is $\operatorname{rank} A = \dim \operatorname{Col} A$. By the line above, $\operatorname{rank} A = \dim \operatorname{Row} A = \operatorname{rank} A^T$.

Theorem (Rank theorem)
For an $m \times n$ matrix A: $\operatorname{rank} A + \dim \operatorname{Nul} A = n$.

Proof. Every column of A is either a pivot column (counted by $\operatorname{rank} A$) or corresponds to a free variable (counted by $\dim \operatorname{Nul} A$). There are n columns. □

- Example: a 7×9 matrix with a two-dimensional null space has rank 7, so $\operatorname{Col} A$ is a 7-dimensional subspace of $\mathbb{R}^7$, i.e. all of $\mathbb{R}^7$.
- A 6×9 matrix has rank at most 6, so its null space has dimension at least 3.

More statements for the invertible matrix theorem (A is $n \times n$): (m) the columns of A form a basis of $\mathbb{R}^n$; (n) $\operatorname{Col} A = \mathbb{R}^n$; (o) $\operatorname{rank} A = n$; (p) $\dim \operatorname{Nul} A = 0$; (q) $\operatorname{Nul} A = \{\mathbf{0}\}$.

Shortcuts

- **Bases for the three spaces from one row reduction**: $\operatorname{Col} A$: pivot columns of A; $\operatorname{Row} A$: nonzero rows of the echelon form; $\operatorname{Nul} A$: vectors from the parametric form.
- **Counting**: $\operatorname{rank} A = \#\text{pivots} = \dim \operatorname{Col} A = \dim \operatorname{Row} A$; $\dim \operatorname{Nul} A = n - \operatorname{rank} A$.
- **Is this set a basis of $\mathbb{R}^n$?** With exactly n vectors, check independence only (or $\det \neq 0$). With a different number of vectors: no.
- **Basis of a subspace given as a span**: put the vectors as columns and keep the pivot columns.
- **Extending to a basis of $\mathbb{R}^n$**: row reduce $[\,\mathbf{v}_1 \cdots \mathbf{v}_k\ \mathbf{e}_1 \cdots \mathbf{e}_n\,]$ and keep the pivot columns; the $\mathbf{v}$'s come first, so they are all kept.
- **Polynomial subspaces**: write the conditions as equations on the coefficients; solve; read off a basis.
- **Rank questions**: $\operatorname{rank} A \leq \min(m, n)$. Use the rank theorem before computing anything.

Traps

- The basis for $\operatorname{Col} A$ is the pivot columns *of* A. The columns of the echelon form span a different space in general.
- The basis for $\operatorname{Row} A$ is rows *of the echelon form*; the corresponding rows of A need not work (row swaps change which rows they are).
- A basis is not unique; its size is.
- $\dim \mathbb{R}^2 = 2$, but $\mathbb{R}^2$ is not a subspace of $\mathbb{R}^3$; a plane through $\mathbf{0}$ in $\mathbb{R}^3$ is a 2-dimensional subspace.
- The rank theorem has n (columns), not m.

§3 Homework this week

HW5, Section 4.3: 3, 5, 13, 15, 20, 23, 24, 26, 29, 31, 41, 44. HW6, Section 4.5: 1, 3, 9, 11, 17, 18, 22, 24, 25, 27, 33, 35, 44, 45 (all the true/false questions in 4.5 are recommended). (Section 4.4 of HW6 is in the Week 7 handout.) Midterm 1 is this week.

§4 Problem set

P1. For

$$A = \begin{bmatrix} -2 & 4 & -2 & -4 \\ 2 & -6 & -3 & 1 \\ -3 & 8 & 2 & -3 \end{bmatrix},$$

find bases for $\operatorname{Col} A$, $\operatorname{Row} A$ and $\operatorname{Nul} A$, and check the rank theorem.

P2. Is $\{(1, 0, -1), (2, 1, 0), (1, 3, 4)\}$ a basis of $\mathbb{R}^3$?

P3. Find a basis for, and the dimension of, $H = \operatorname{Span}\{(1, 0, 2), (2, 1, 5), (0, -1, -1)\}$.

P4. Find a basis for $\{p \in \mathbb{P}_3 : p(1) = 0\}$ and its dimension.

P5. (a) A is 5×8 with rank 5. What is $\dim \operatorname{Nul} A$? Is $\operatorname{Col} A = \mathbb{R}^5$? (b) Can a 4×6 matrix have a one-dimensional null space?

P6. Extend $\{(1,2,0),(0,1,3)\}$ to a basis of $\mathbb{R}^3$.

P7. True or false, with a reason. (a) If $\dim V = 3$, any 4 vectors in V are dependent. (b) If $\dim V = 3$, any 2 vectors in V span a plane. (c) The number of pivot columns of A equals $\dim \operatorname{Row} A$. (d) If A is 3×3 and $\dim \operatorname{Nul} A = 1$, then $\det A = 0$.